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Savitzky-Golay split command-lag search

rotor median publish period [s]: 0.00495767593384
gimbal median publish period [s]: 0.00495719909668
initial lags [s]: {'rotor': 0.004957675933837891, 'gimbal': 0.004957199096679688}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.0555392062s gimbal=0.00797288198s cost=583.287350721
    lag_gradient={'rotor': -4.299962992948547e-06, 'gimbal': 1.1255722268677104e-06}
    FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 41.49403053843453, 'finite_difference_l2_norm': 41.494030537216,
    'difference_l2_norm': 1.3883058570554929e-07, 'relative_error': 3.345796585775276e-09},
    {'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 3.667091584470166, 'finite_difference_l2_norm': 3.667091504200583,
    'difference_l2_norm': 2.427589881252419e-07, 'relative_error': 6.619932514183895e-08},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 6.617745343528772, 'finite_difference_l2_norm': 6.617745241169013,
    'difference_l2_norm': 3.4275172799411724e-07, 'relative_error': 5.179282523001288e-08}]
smooth width=2: rotor=0.0562485627s gimbal=0.00266503168s cost=583.272969983
    lag_gradient={'rotor': 2.736420724558375e-05, 'gimbal': 4.733120853850359e-05}
    FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 16.00058038137841, 'finite_difference_l2_norm': 16.00058017245243,
    'difference_l2_norm': 1.7456365875336017e-06, 'relative_error': 1.0909832930592856e-07},
    {'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 4.60790425602606, 'finite_difference_l2_norm': 4.6079035797049,
    'difference_l2_norm': 1.2734273022493537e-06, 'relative_error': 2.763571531643716e-07},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 7.728564203639891, 'finite_difference_l2_norm': 7.728563415895859,
    'difference_l2_norm': 2.7214318639613387e-06, 'relative_error': 3.52126448361474e-07}]
smooth width=1: rotor=0.0561974939s gimbal=0.00638762191s cost=583.267876276
    lag_gradient={'rotor': 0.000427089873929698, 'gimbal': -0.35797669173849234}
    FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 44.79275405373648, 'finite_difference_l2_norm': 44.79085344738733,
    'difference_l2_norm': 0.4374133407596534, 'relative_error': 0.00976527007548815},
    {'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 6.729746168966109, 'finite_difference_l2_norm': 6.72974072561678,
    'difference_l2_norm': 7.532426362849569e-06, 'relative_error': 1.1192734723911252e-06},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 9.27228670772365, 'finite_difference_l2_norm': 9.291144541654514,
    'difference_l2_norm': 0.45162477837871523, 'relative_error': 0.04860808874019438}]
smooth width=0.5: rotor=0.0560638216s gimbal=0.00644269036s cost=583.260522401
    lag_gradient={'rotor': 0.021941612950229894, 'gimbal': -1.2878923917154208}
    FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 15.553694122400204, 'finite_difference_l2_norm': 15.550139251670682,
    'difference_l2_norm': 0.8230424656729305, 'relative_error': 0.0529162049347233},
    {'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 9.163050832070944, 'finite_difference_l2_norm': 9.163029506819925,
    'difference_l2_norm': 3.3330312079664656e-05, 'relative_error': 3.6374688616817005e-06},
    {'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
    'analytic_l2_norm': 10.665853069836489, 'finite_difference_l2_norm': 10.661550944221183,
    'difference_l2_norm': 0.8248745984050435, 'relative_error': 0.07733789252524263}]

strict profile iterations: 1
iteration 1: best=(0.05606382158, 0.006442690363) cost=583.25746826 rotor_range=(0.051106145646,
     0.061021497514) gimbal_range=(0.001485491267, 0.01139988946) expansions=[]
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selected rotor lag 0.05606382158 s (11.3085 publish periods)
selected gimbal lag 0.006442690363 s (1.29966 publish periods)
selected objective cost 583.257467614
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